Program : **Diploma in Communication and Computer Networking**	
Course Code : **4322**	Course Title: **Computer Networks**
Semester : **4**	Credits: **4**
Course Category: **Programme core course**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- Understand the fundamentals of networking and the topologies
- Understand LAN Components and Protocols.
- Understand Network Addressing.
- Understand about Internet

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Describe the fundamentals of Networking and Topologies	10	Understanding
CO2	To interpret LAN Components and Protocols	12	Applying
CO3	To explain Network Addressing.	13	Applying
CO4	To describe the Internet and its service providers	10	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	3			1		
CO2	3	3			2		
CO3	3	3			1		
CO4	3	3			1		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe the fundamentals of Networking and Topologies**		
M1.01	To understand the Overview of Networking	3	Understanding
M1.02	To Understand Network Communication Standards	3	Understanding
M1.03	To get an Overview of network topologies	4	Understanding
Contents: Introduction to Networking and Topologies Overview of Networking--Definition & history of networks -Usage of Computer Networks-Standard Organizations and Protocols-Hardware and Software components Network Communication Standards-OSI Reference Model-TCP/IP Model-Compare and contrast OSI and TCP/IP models Overview of network topologies- Basic topologies- bus, ring, star, mesh and hybrid.			
CO2	**To interpret LAN Components and Protocols.**		
M2.01	To learn about LAN Cables and Connectors	3	Understanding
M2.02	To learn about LAN devices	4	Understanding
M2.03	To learn about lower layer protocols	5	Understanding
	Series Test –I	1	
Contents: LAN Components and Protocols : LAN Cables and connectors – Co-axial, twisted pair, optical fiber. LAN devices – repeaters, hubs, switches, NIC, WLANs. Overview of Lower Layer Protocols – ARC net, Ethernet, Ethernet Communication, Fast Ethernet, Gigabit Ethernet, Token Ring, Token Ring Frame format, Fault Management and tolerance, FDDI. Wireless LAN- Introduction-Fundamentals of WLAN-IEEE 802.11-Bluetooth			
CO3	**To explain Network Addressing.**		
M3.01	To Identify the features of the different IP address classes.	3	Understanding

M3.02	To Recognize the concepts of subnetting	4	Applying
M3.03	To understand the difference between IPV4 and IPV6	5	Understanding
Contents: **Network Addressing:** Introduction-TCP/IP addressing scheme-Components of IP addressing-IP address classes Limitations of IP address. IP subnetting –Creating subnets in networks-Communication across subnets –Subnetting Considerations-Subnetting Limitations-IPV4, IPv6			
CO4	**To describe the Internet and its Service Providers**		
M 4.01	To understand Internet and connectivity	5	Understanding
M 4.02	To understand ISP Web hosting	5	Understanding
	Series Test – II	1	
Contents: **Internet and its Service Providers:** Internet-connection types- ISP Web hosting-Top Web Hosting Companies IANA- IANA Root Zone Database IANA Number -Resources Local Internet registry (LIR)-National Internet Registry (NIR)-AfriNIC, APNIC, ARIN, LACNIC, RIPE NCC Regional Internet Registry (RIR)-Registration of a domain- Top Domain Registrars-Registrar for . EDU.IN, .RES.IN, .AC.IN, .GOV.IN in INDIA			

Text / Reference

T/R	Book Title/Author
T1	Data Communications and Networking - Behrouz A Forouzan, Tata McGraw-Hill, 5th edition, ISBN: 9780070634145 for Unit I and III.
T2	Basics of Networking, PHI learning Pvt. Ltd. 2013, ISBN: 978-81-203-2489 for Unit II, IV, V and VI.
T3	Computer Networks is an introductory book written by. Andrew S. Tanenbaum
T4	Data & Computer Communication, Williams Stallings, Prentice Hall of India
R1	Information Technology Today ,S. Jaiswal, Galgotia Publications
R2	Computer Networks , Bhushan Trivedi , Oxford University Press, 2013
R3	Data Communication & Networking, Forouzen ,Tata McGraw Hill

R4	Networks for Computer Scientists and Engineers ,Youlu Zheng & Shakil Akhtar, Oxford University Press, 2012

Online Resources

Sl.No	Website Link
1 2	http://nptel.iitm.ac.in/courses.php?disciplineId=106 ii. http://www.edrawsoft.com